

The impact of tropospheric versus stratospheric CO₂ forcing on the Hadley cell

Climate physics seminar

Ole Merkes

Advised by Sarah Kang and Moritz Günther

August 17, 2026



MAX-PLANCK-INSTITUT
FÜR METEOROLOGIE



Universität Hamburg
DER FORSCHUNG | DER LEHRE | DER BILDUNG

1 Introduction

2 Results

3 Analysis

Hadely cell (HC) trends under greenhouse gas forcing

Projections from CMIP studies

- HC **weakening** (Lu et al., 2007; Xia et al., 2020)
- HC **widening** at the poleward edge (Lu et al., 2007; Xia et al., 2020)

Hadely cell (HC) trends under greenhouse gas forcing

Projections from CMIP studies

- HC **weakening** (Lu et al., 2007; Xia et al., 2020)
- HC **widening** at the poleward edge (Lu et al., 2007; Xia et al., 2020)

How does the HC respond to different vertical forcing structures?

Motivation for experimental setup

- Investigation of different meridional forcing structures (Kim et al., 2022; Shaw & Tan, 2018)

Motivation for experimental setup

- Investigation of different meridional forcing structures (Kim et al., 2022; Shaw & Tan, 2018)
- **Mechanisms** explaining weakening are related to tropospheric parameters (tropopause height, static stability, meridional temperature gradient) (Chemke & Polvani, 2021)

Motivation for experimental setup

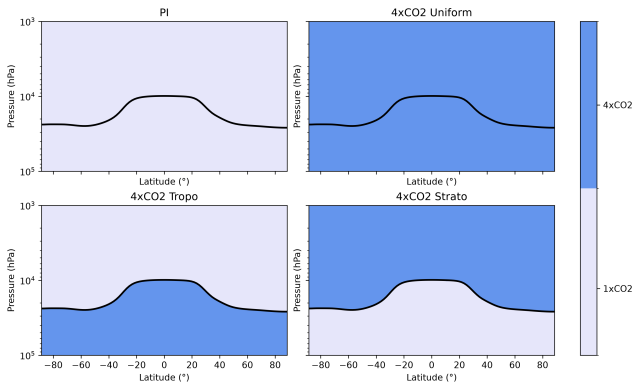
- Investigation of different meridional forcing structures (Kim et al., 2022; Shaw & Tan, 2018)
- **Mechanisms** explaining weakening are related to tropospheric parameters (tropopause height, static stability, meridional temperature gradient) (Chemke & Polvani, 2021)
- **Stratospheric ozone** historically lead to a strengthening of the HC (Son et al., 2010)

Motivation for experimental setup

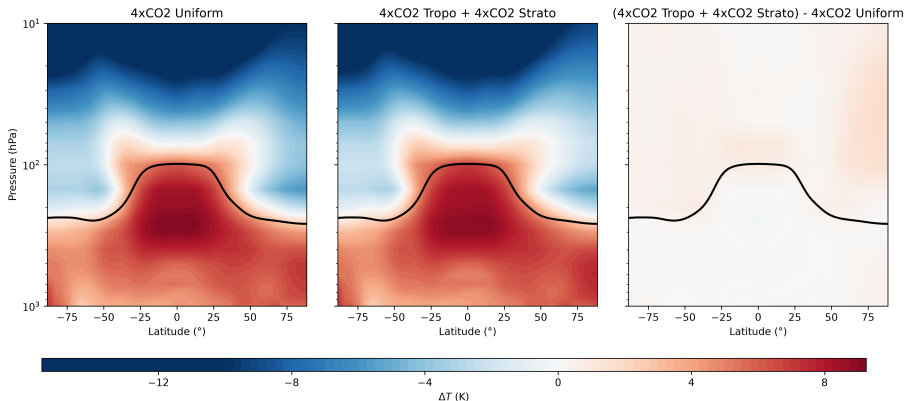
- Investigation of different meridional forcing structures (Kim et al., 2022; Shaw & Tan, 2018)
- **Mechanisms** explaining weakening are related to tropospheric parameters (tropopause height, static stability, meridional temperature gradient) (Chemke & Polvani, 2021)
- **Stratospheric ozone** historically lead to a strengthening of the HC (Son et al., 2010)
- Different **adjustment time** scales for stratosphere and troposphere

Experimental setup

- ECHAM6 is a CMIP6 type general circulation model
- Slab-ocean ($d = 50$ m) configuration



Verifying decomposition



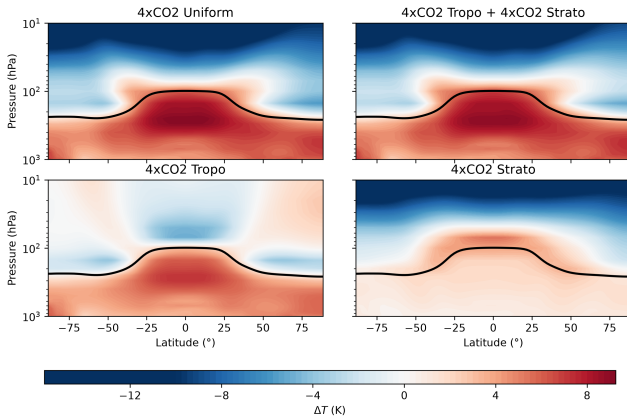
The temperature response of our forcings is **linear**.

1 Introduction

2 Results

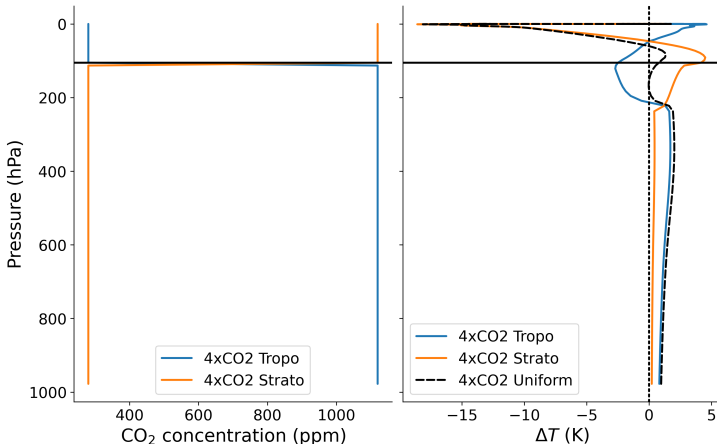
3 Analysis

Temperature anomalies



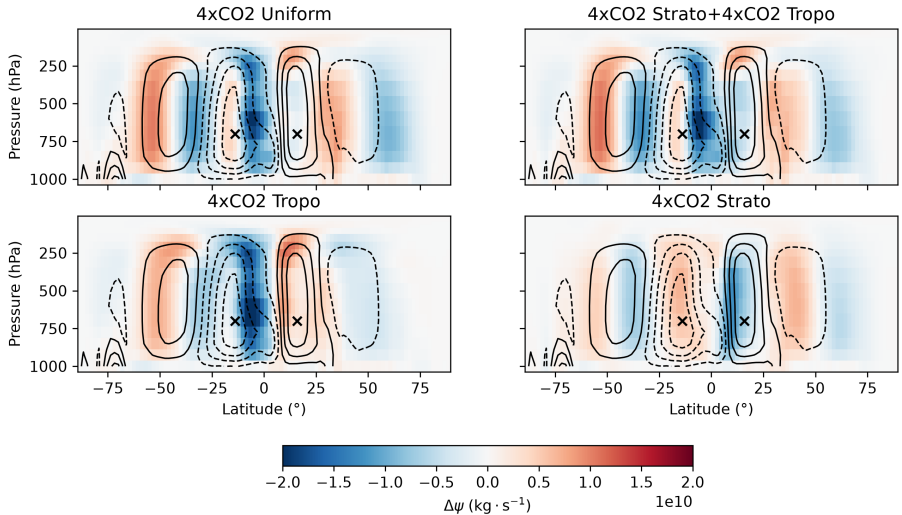
- Tropospheric warming in 4xCO₂ Tropo
- Stratospheric cooling in 4xCO₂ Strato

RCE model



Heating/cooling above the tropopause is due to changes in optical thickness

Zonally averaged circulation



Changes in Hadley circulation

	Uniform	Tropo	Strato
$\Delta\psi_{\max,\text{NH}}$	-1.9%	3.6%	-8.9%
$\Delta\Phi_{\text{H,NH}}$	1.6°	0.7°	0.7°

Unexpected response

- HC is **weakening** with stratospheric CO₂
- HC is **strengthening** with tropospheric CO₂

New research question

Why is the HC weakening with stratospheric CO₂ forcing but strengthening with tropospheric CO₂ forcing?

New research question

Why is the HC weakening with stratospheric CO₂ forcing but strengthening with tropospheric CO₂ forcing?

Methodology

Employ energy budget framework of the atmosphere to explain this behavior.

1 Introduction

2 Results

3 Analysis

The energetic framework

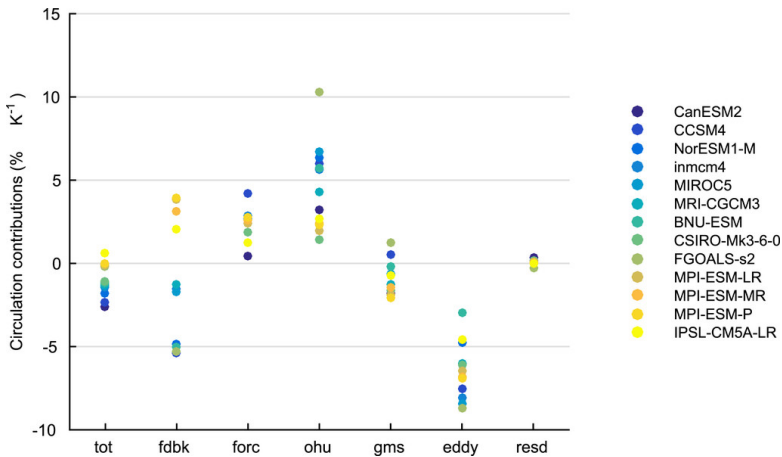
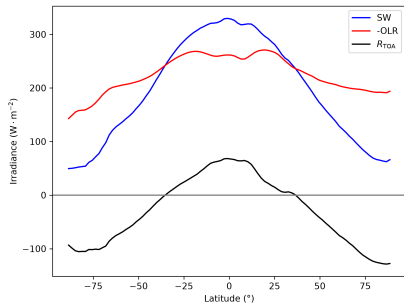


Figure: Contributions to changes in Hadley circulation for multiple models (Feldt & Bordoni, 2016).

Energy transport

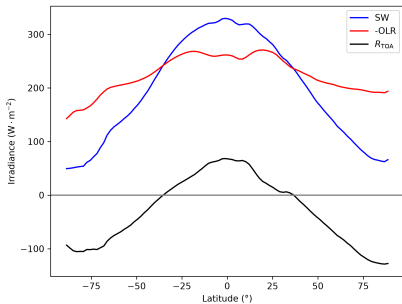


Energy transport

Flux to transport

$$R_{\text{TOA}} = \frac{\partial F}{\partial y}$$

$$F_{ao} = \int_{-\pi/2}^{\phi} \int_0^{2\pi} R_{\text{TOA}} a^2 \cos \phi d\lambda d\phi$$



Energy transport by the Hadley cell?

Linearisation

$$F_{\text{HC}} = \psi_{\text{max}} H$$

$$H \sim \frac{1}{\psi_{\text{max}}} \int_0^{p_s} [\bar{v}][\bar{h}] dp$$

$$\frac{\delta \psi_{\text{max}}}{\psi_{\text{max}}} = \frac{\delta F_{\text{HC}}}{F_{\text{HC}}} - \frac{\delta H}{H}$$

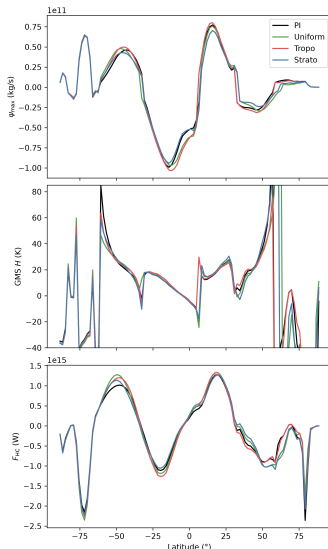
Energy transport by the Hadley cell?

Linearisation

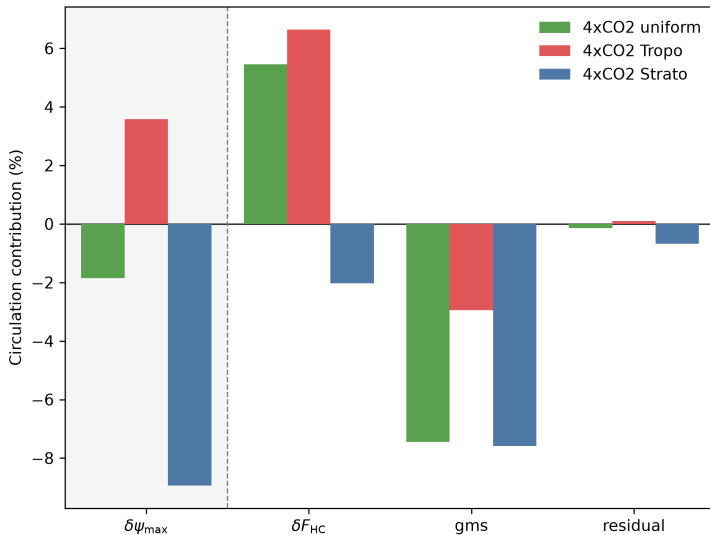
$$F_{\text{HC}} = \psi_{\text{max}} H$$

$$H \sim \frac{1}{\psi_{\text{max}}} \int_0^{p_s} [\bar{v}][\bar{h}] dp$$

$$\frac{\delta \psi_{\text{max}}}{\psi_{\text{max}}} = \frac{\delta F_{\text{HC}}}{F_{\text{HC}}} - \frac{\delta H}{H}$$



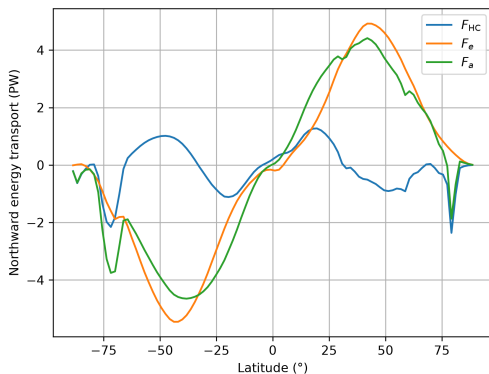
GMS weakens the circulation



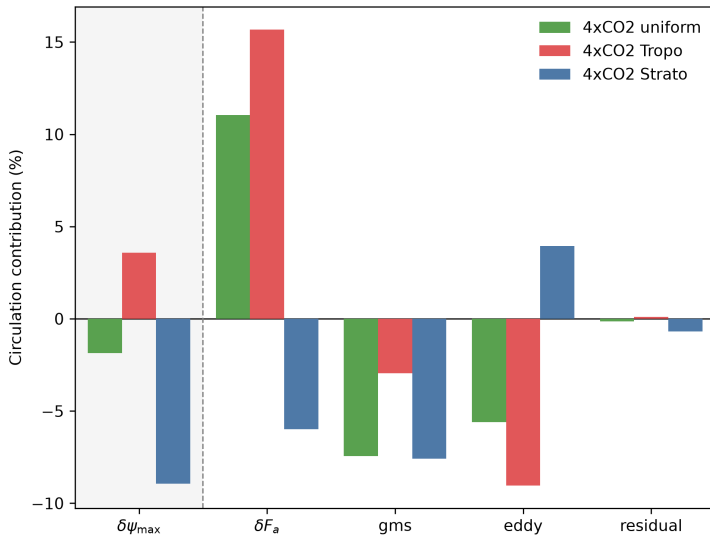
Eddy energy transport

Linearisation

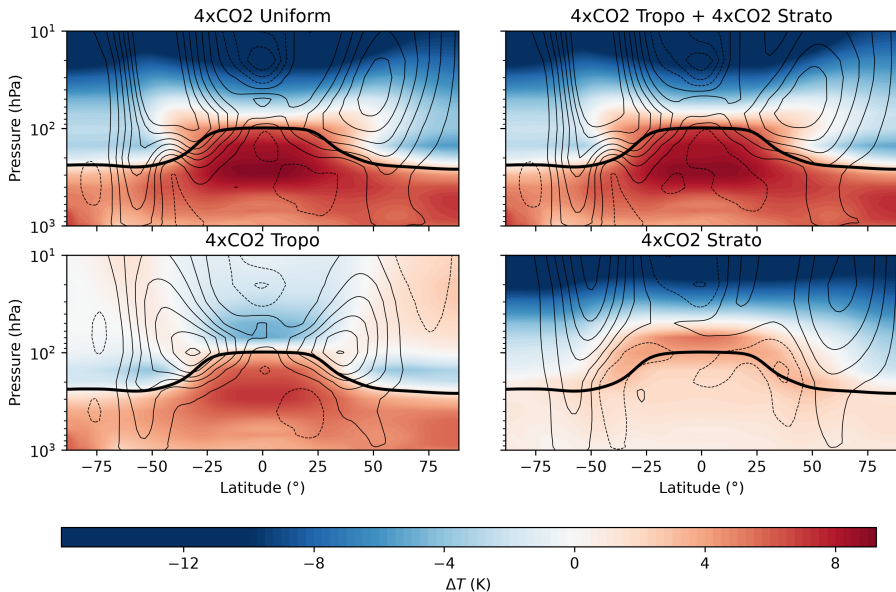
$$F_{\text{HC}} = F_a - F_e$$
$$\frac{\delta\psi_{\text{max}}}{\psi_{\text{max}}} = \frac{\delta F_a}{F_{\text{HC}}} - \frac{\delta F_e}{F_{\text{HC}}} - \frac{\delta H}{H}$$



Opping eddy contributions



Meridional temperature gradient and jet strength



Forcing and feedbacks

Linearisation

$$\Delta R_{\text{TOA}} = \sum_i \lambda_i \Delta T_s + R_f$$

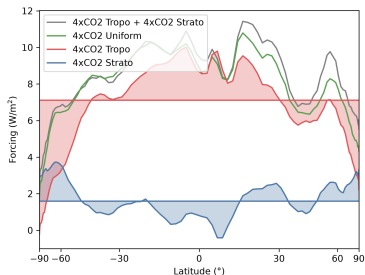
$$\frac{\delta \psi_{\text{max}}}{\psi_{\text{max}}} = \frac{\delta F_{\text{forc}}}{F_{\text{HC}}} + \frac{\delta F_{\text{fdb}}}{F_{\text{HC}}} - \frac{\delta F_{\text{e}}}{F_{\text{HC}}} - \frac{\delta H}{H}$$

Forcing and feedbacks

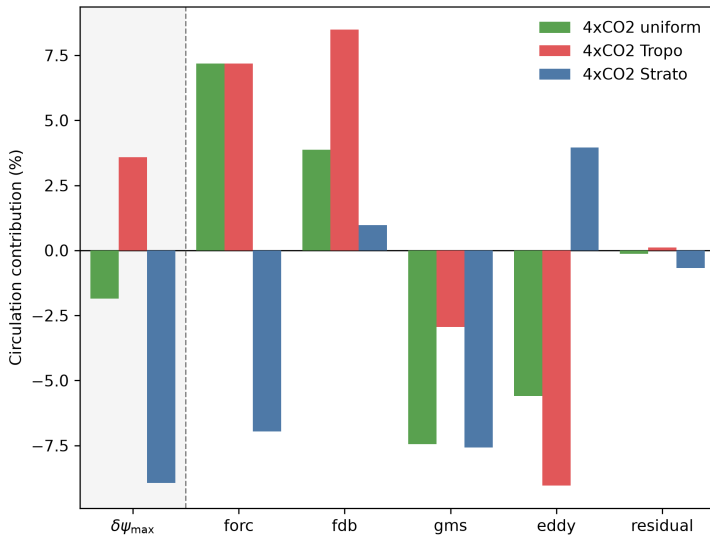
Linearisation

$$\Delta R_{\text{TOA}} = \sum_i \lambda_i \Delta T_s + R_f$$

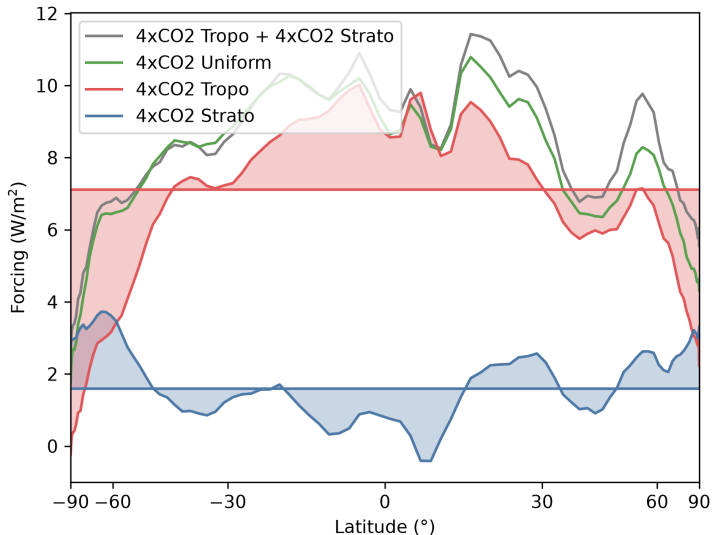
$$\frac{\delta \psi_{\text{max}}}{\psi_{\text{max}}} = \frac{\delta F_{\text{forc}}}{F_{\text{HC}}} + \frac{\delta F_{\text{fdb}}}{F_{\text{HC}}} - \frac{\delta F_e}{F_{\text{HC}}} - \frac{\delta H}{H}$$



Forcing contributions



Meridional forcing structure



Summary

Research question

How does the atmosphere respond to forcing in the troposphere compared to the stratosphere?

Summary

Research question

How does the atmosphere respond to forcing in the troposphere compared to the stratosphere?

Key takeaways

- Tropospheric and stratospheric CO₂ contribute very differently to Hadley circulation changes.

Summary

Research question

How does the atmosphere respond to forcing in the troposphere compared to the stratosphere?

Key takeaways

- Tropospheric and stratospheric CO_2 contribute very differently to Hadley circulation changes.
- HC is strengthening with $4\times\text{CO}_2$ in the troposphere and weakening with $4\times\text{CO}_2$ in the stratosphere.

Summary

Research question

How does the atmosphere respond to forcing in the troposphere compared to the stratosphere?

Key takeaways

- Tropospheric and stratospheric CO₂ contribute very differently to Hadley circulation changes.
- HC is strengthening with 4xCO₂ in the troposphere and weakening with 4xCO₂ in the stratosphere.
- Circulation changes can be attributed to changes in forcing, feedback, GMS, and eddies.

Summary

Research question

How does the atmosphere respond to forcing in the troposphere compared to the stratosphere?

Key takeaways

- Tropospheric and stratospheric CO_2 contribute very differently to Hadley circulation changes.
- HC is strengthening with $4\times\text{CO}_2$ in the troposphere and weakening with $4\times\text{CO}_2$ in the stratosphere.
- Circulation changes can be attributed to changes in forcing, feedback, GMS, and eddies.
- Meridional forcing structure as the main driver for the different circulation response.

Summary

Research question

How does the atmosphere respond to forcing in the troposphere compared to the stratosphere?

Key takeaways

- Tropospheric and stratospheric CO₂ contribute very differently to Hadley circulation changes.
- HC is strengthening with 4xCO₂ in the troposphere and weakening with 4xCO₂ in the stratosphere.
- Circulation changes can be attributed to changes in forcing, feedback, GMS, and eddies.
- Meridional forcing structure as the main driver for the different circulation response.

Thank you for your attention!

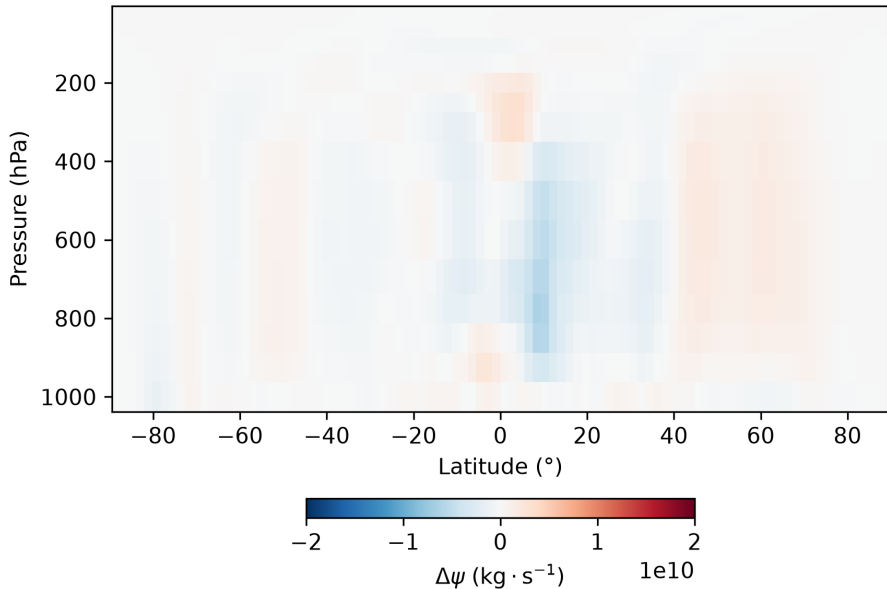
References I

- Chemke, R., & Polvani, L. M. (2021). Elucidating the Mechanisms Responsible for Hadley Cell Weakening Under $4 \times \text{CO}_2$ Forcing. *Geophysical Research Letters*, 48(3), e2020GL090348. <https://doi.org/10.1029/2020GL090348>
- Feldl, N., & Bordoni, S. (2016). Characterizing the Hadley Circulation Response through Regional Climate Feedbacks. *Journal of Climate*, 29(2), 613–622. <https://doi.org/10.1175/JCLI-D-15-0424.1>
- Kim, D., Kim, H., Kang, S. M., Stuecker, M. F., & Merlis, T. M. (2022). Weak Hadley cell intensity changes due to compensating effects of tropical and extratropical radiative forcing. *npj Climate and Atmospheric Science*, 5(1), 61. <https://doi.org/10.1038/s41612-022-00287-x>
- Lu, J., Vecchi, G. A., & Reichler, T. (2007). Expansion of the Hadley cell under global warming. *Geophysical Research Letters*, 34(6), 2006GL028443. <https://doi.org/10.1029/2006GL028443>

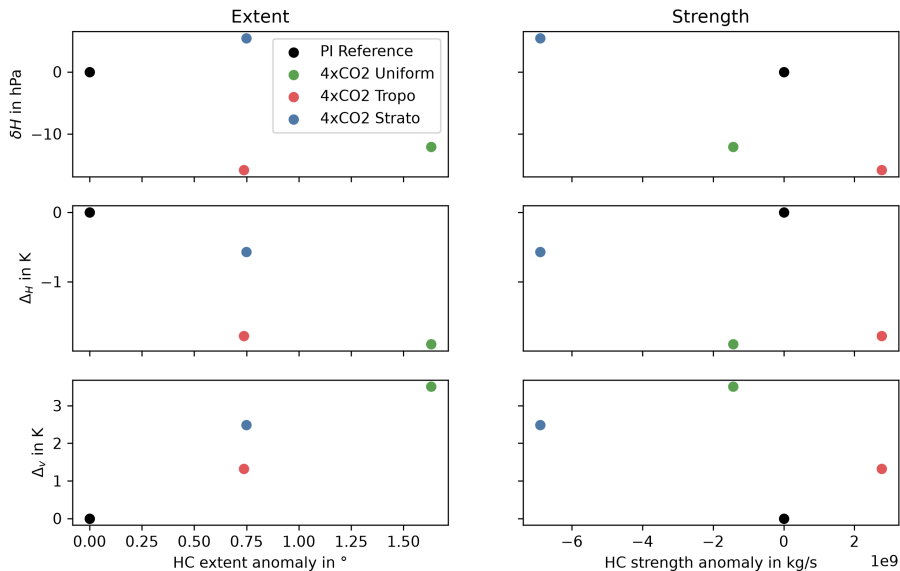
References II

- Shaw, T. A., & Tan, Z. (2018). Testing Latitudinally Dependent Explanations of the Circulation Response to Increased CO₂ Using Aquaplanet Models. *Geophysical Research Letters*, 45(18), 9861–9869. <https://doi.org/10.1029/2018GL078974>
- Son, S.-W., Gerber, E. P., Perlwitz, J., Polvani, L. M., Gillett, N. P., Seo, K.-H., Eyring, V., Shepherd, T. G., Waugh, D., Akiyoshi, H., Austin, J., Baumgaertner, A., Bekki, S., Braesicke, P., Brühl, C., Butchart, N., Chipperfield, M. P., Cugnet, D., Dameris, M., ... Yamashita, Y. (2010). Impact of stratospheric ozone on Southern Hemisphere circulation change: A multimodel assessment. *Journal of Geophysical Research: Atmospheres*, 115(D3). <https://doi.org/10.1029/2010JD014271>
- Xia, Y., Hu, Y., & Liu, J. (2020). Comparison of trends in the Hadley circulation between CMIP6 and CMIP5. *Science Bulletin*, 65(19), 1667–1674. <https://doi.org/10.1016/j.scib.2020.06.011>

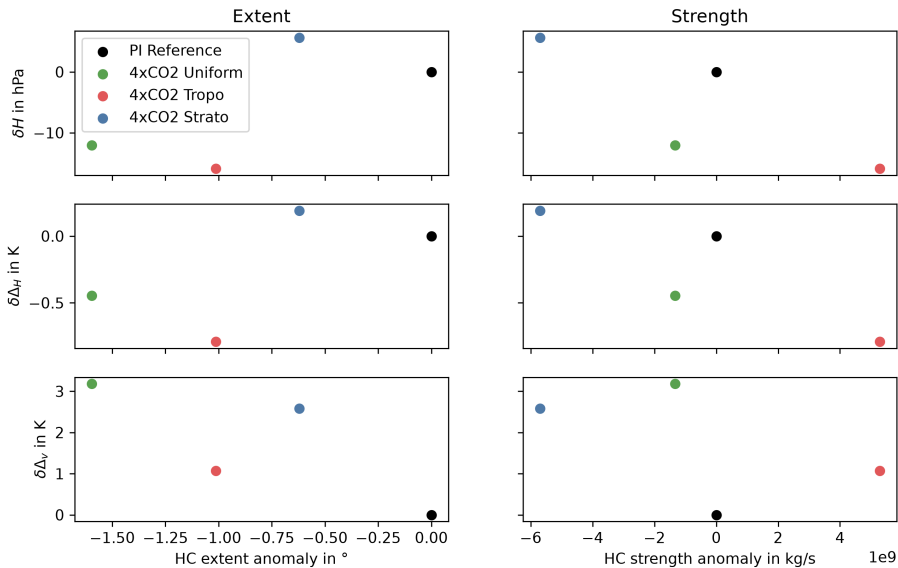
4xCO₂ Tropo + 4xCO₂ Strato - 4xCO₂ Uniform



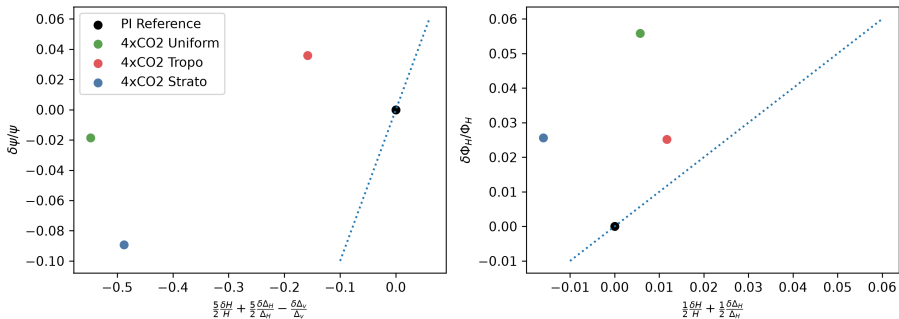
Northern Hemisphere



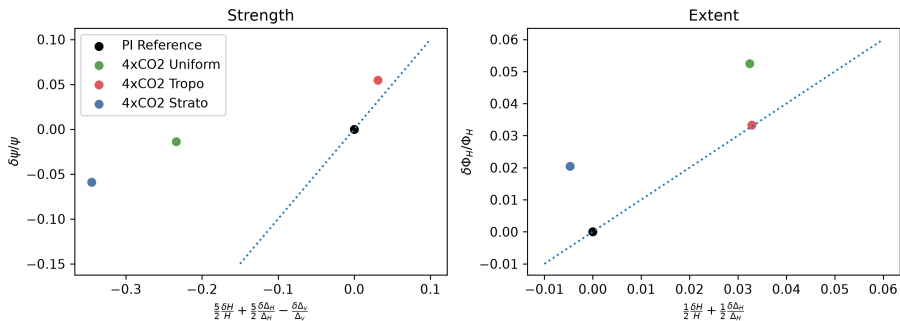
Southern Hemisphere



Northern hemisphere

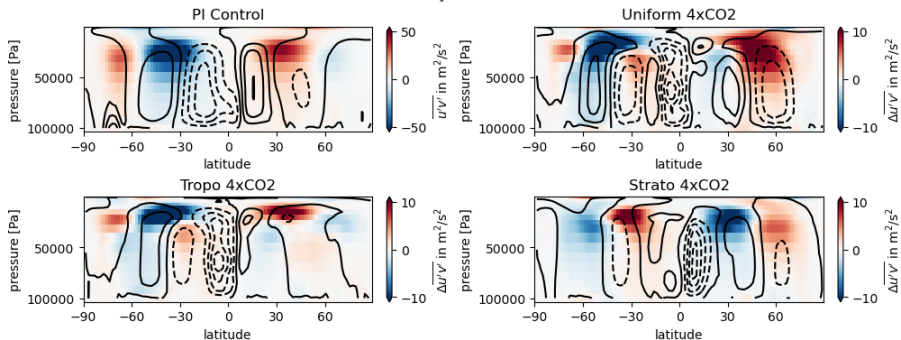


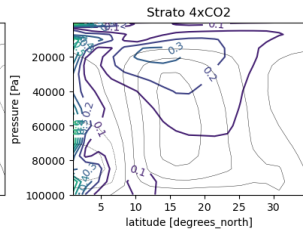
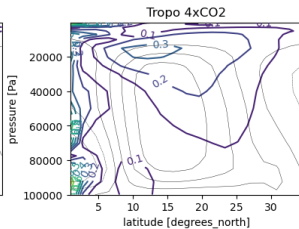
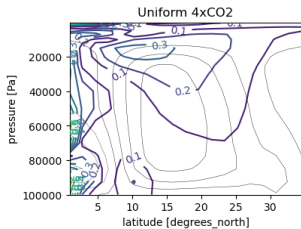
Southern Hemisphere



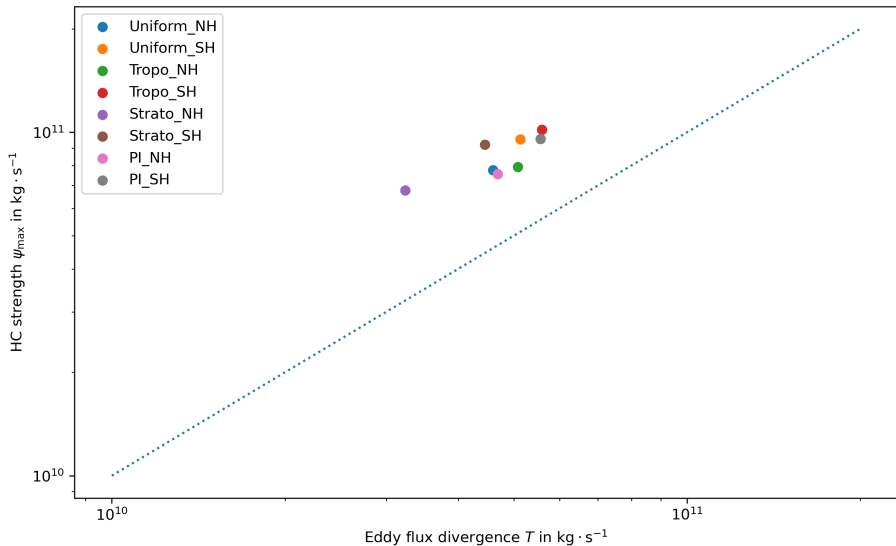
Eddy anomalies

Anomalous eddy momentum fluxes

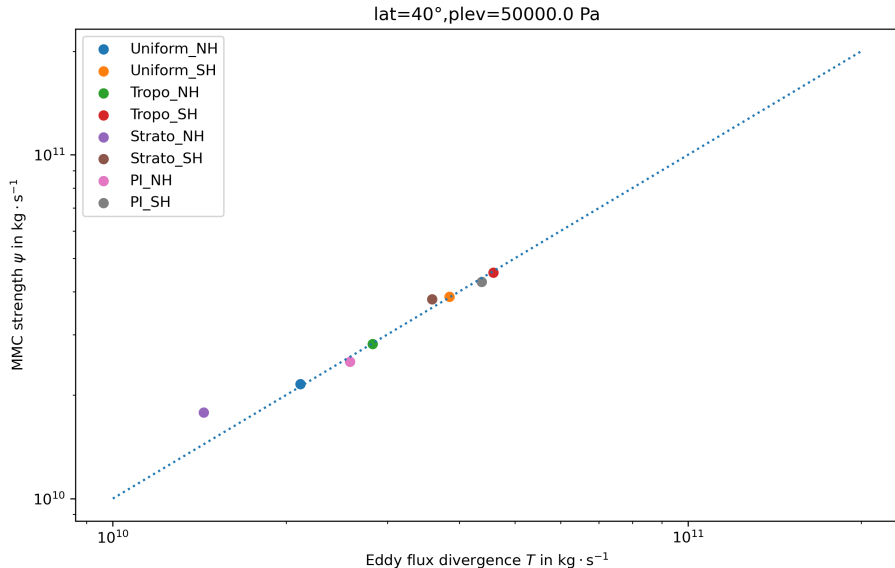




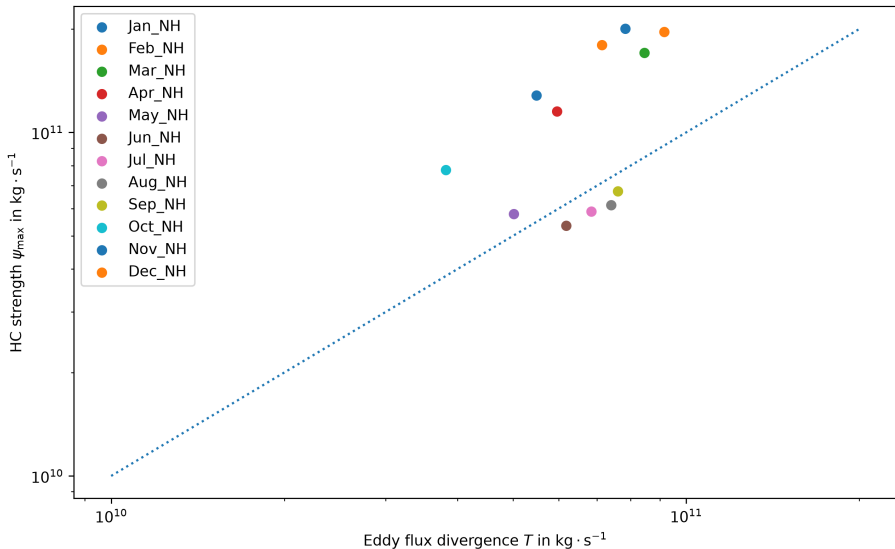
HC strength versus eddy driven HC strength



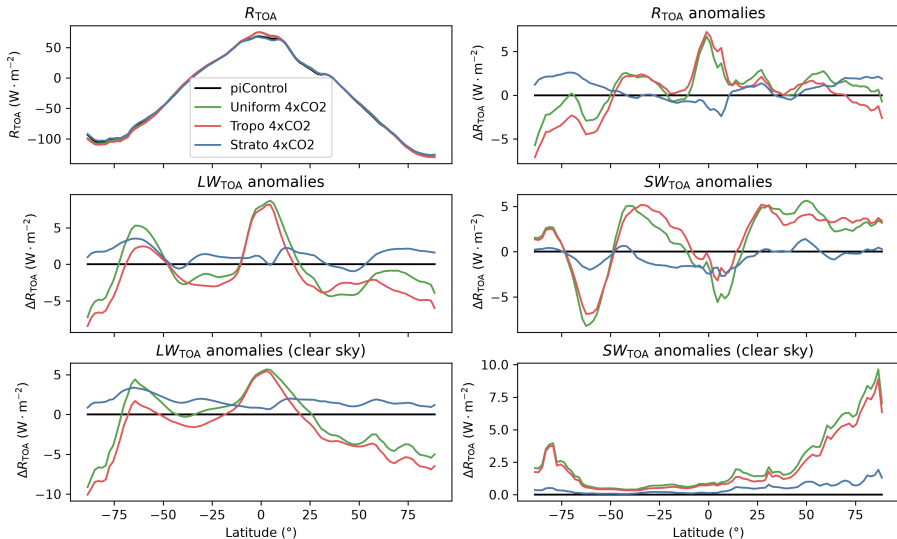
Ferrel cell strength versus eddy driven Ferrel cell strength



Seasonality



SW LW clear sky



Sea ice (ocean transport)

